

READING PASSAGE 2

The Development of Quantum Computing

A The study of the development of quantum computing has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The intellectual lineage of this field extends considerably further than many realise. While modern research techniques have only recently been applied to these questions, philosophers and natural scientists have grappled with related issues for centuries. The transition from speculative philosophical inquiry to systematic empirical investigation represents one of the most significant developments in the intellectual history of this domain.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D A substantial body of empirical evidence supports these theoretical frameworks. A landmark study conducted by researchers at leading international institutions analysed data spanning a thirty-year period and involving participants from more than forty countries. The findings revealed a consistent pattern of results that held across diverse cultural and economic contexts, suggesting that the identified mechanisms operate with considerable universality. The magnitude of the observed effects was particularly striking, with the key variables accounting for a substantial proportion of the total variance in outcomes.

E Despite the substantial progress that has been made, significant challenges and controversies remain. Critics have raised important methodological concerns regarding the generalisability of findings derived from studies conducted in specific contexts. The predominance of research carried out in Western, educated, industrialised, rich, and democratic societies raises questions about the applicability of current theories to populations with different cultural, economic, and political circumstances. Furthermore, the reliance on cross-sectional designs in much of the existing literature limits the capacity to draw strong causal inferences.

F The implications of this body of research extend well beyond the academy. As societies grapple with increasingly complex challenges, the insights generated by scholarly inquiry in this domain have the potential to inform more effective and equitable policy responses. Realising this potential, however, will require sustained investment in both fundamental research and the translational infrastructure needed to convert knowledge into action.

G In conclusion, the study of the development of quantum computing represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Question 1

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Question 2

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 2 The author believes that all research questions in this field have been answered.

Question 3

Choose the correct letter, A, B, C or D.

- 3 The meta-analyses cited in the passage reveal that:
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone

Question 4

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 4 The landmark study analysed data spanning a period of _____ years.

Question 5

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 5 All critics agree that the research findings are universally applicable.

Question 6

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 6 Researchers used data from more than _____ countries in the landmark study.

Questions 7–8

Choose the correct letter, A, B, C or D.

- 7 What finding emerged from the landmark study discussed in the passage?
- A. excessive research funding

- B. translating academic findings into practical policy
- C. lack of public interest
- D. too many international collaborations

8 The passage indicates that critics have raised concerns about:

- A. excessive research funding
- B. translating academic findings into practical policy
- C. lack of public interest
- D. too many international collaborations

Question 9

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 9 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Questions 10–11

Choose the correct letter, A, B, C or D.

10 What does the passage identify as a significant challenge to current understanding?

- A. minimal impact
- B. creating underlying conditions for action
- C. determining individual choices directly
- D. irrelevant to outcomes

11 Which of the following best describes the author's view on future developments?

- A. excessive funding
- B. theoretical and methodological limitations
- C. public interest
- D. international collaboration

Question 12

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 12 The landmark study discussed in the passage involved participants from multiple countries.

Question 13

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

13 Case studies have documented the effects of _____ change on the system.